

2. Lidocaine is a local anaesthetic used by dentists to numb pain. It is believed that it blocks the sodium ion channels in a neurone membrane.

(a) Describe the function of sodium ion channels in a neurone membrane in response to a stimulus. [2]

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(b) Explain how lidocaine acts as a local anaesthetic. [4]

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The maximum allowable dose of lidocaine for a patient is 7 mg per kg of tissue (mg kg<sup>-1</sup>).

(c) Use the equation below to calculate the maximum dose volume for a 60 kg patient if the concentration of lidocaine is 2%. [2]

maximum dose volume (cm<sup>3</sup>) =  $\frac{\text{maximum allowable dose (mg kg}^{-1}\text{)}}{\text{concentration of lidocaine (\%)}}$   $\times \frac{\text{mass of patient (kg)}}{10}$   $\times \frac{1}{\text{concentration of lidocaine (\%)}}$

maximum dose volume = ..... cm<sup>3</sup>



Examiner  
only

(d) The marine cone shell, *Conus magus*, releases a chemical which blocks calcium ion channels on the pre-synaptic membrane of a synapse. This has a potential use as an anaesthetic.

With reference to the **pre-synaptic neurone only**, describe and explain the mechanism by which this chemical could work as an anaesthetic. [3]

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